Department of Electronic Systems

Examination paper for TFE4146 Semiconductor Devices

Examination date: 13.12.2023

Examination time (from-to): 15:00-19:00

Permitted examination support material:
C
Specified printed matter:
- Karl Rottmann, Formelsamling/Collection of mathematical formulas.
- Specified simple calculator

Academic contact during examination: Erik Folven

Phone: 94437094

Academic contact present at the exam location: No

OTHER INFORMATION

Get an overview of the question set before you start answering the questions. Useful formulas, constants and conversions can be found at the end.

Read the questions carefully and make your own assumptions. If a question is unclear/vague, make your own assumptions and specify them in your answer. The academic person is only contacted in case of errors or insufficiencies in the question set. Address an invigilator if you suspect errors or insufficiencies. Write down the question in advance.

Hand drawings: For question 20 you are meant to answer on handwritten sheets. Other questions must be answered directly in Inspera. At the bottom of the question you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit. We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click “Show submission”.

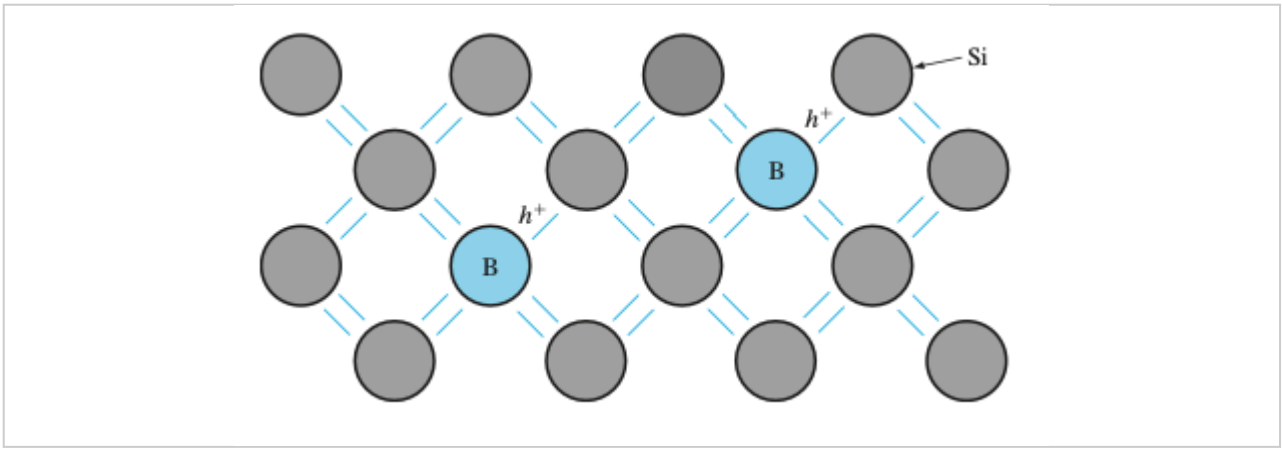
Weighting: The weighting per question is indicated for each question. Blank and wrong answers are typically treated equally in the scoring of multiple choice questions with only one answer, i.e., in these questions there is no risk in guessing an answer if you do not know the correct answer. In questions with more than one alternative there will typically be negative points for wrong/blank answers, but you may not get less than 0p for the question as a whole.

Notifications: If there is a need to send a message to the candidates during the exam (e.g. if there is an error in the question set), this will be done by sending a notification in Inspera. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Access to your answers: After the exam, you can find your answers in the archive in Inspera. Be aware that it may take a working day until any hand-written material is available in the archive.

1



The simple schematic above illustrates a Si crystal at room temperature where some of the Si atoms have been replaced by a B atom. In this case, each of the dopant atoms have an electron less than the Si atoms, introducing an additional hole to the crystal.

How will the introduction of B atoms change the net charge of the crystal?
(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative

- The B doping will not affect the net charge of the crystal
- The B doping will make the crystal negatively charged.
- The B doping will make the crystal positively charged.

What do we typically call the kind of doping shown above?
(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative

- Blood doping
- Acceptor doping
- Donor doping
- Intrinsic doping

How would the introduction of B into a Si crystal be represented in a band diagram representation of the crystal?
(correct: 2p, wrong: 0p, blank: 0p)

Select one alternative

- B doping will increase the band gap of the system.
- The B doping will decrease the conduction band energy.
- B doping will decrease the Fermi energy in the system.
- B doping will increase the mobility of the holes in the system.

Maximum marks: 6

2 Select the correct statements.
(correct: 2p, wrong: -2p, blank: 0p)
Select one or more alternatives:

- The conductivity in a semiconductor is independent of the doping concentration
- In a p-n junction, the conductivity is higher in the neutral region than in the depletion region
- For Si, the mobility is higher for electrons than for holes
- For Si, the lowest conductivity is obtained with zero doping

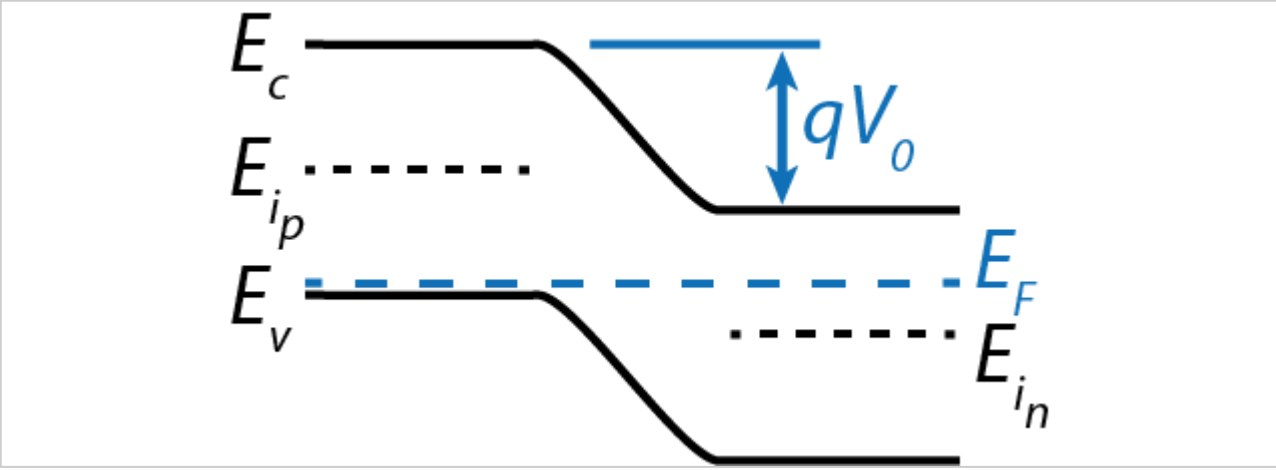
Maximum marks: 4

3 Which of the following experiments/measurements is best suited to determine the mobility of the majority carriers in a uniformly doped semiconductor sample:
(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative:

- the Hall effect experiment
- a C-V measurement
- the Haynes-Shockley experiment
- the Stern–Gerlach experiment

Maximum marks: 2

4



In the following we will consider a Si p-n junction at room temperature. The band diagram at thermal equilibrium for this junction is shown above.

Fill in the blanks in the following two sections of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)

Within the depletion region in the p-type material, there is a net Select alternative (positive, neutral, negative)charge. This region is depleted of free Select alternative (impurities, donors, acceptors, carriers)and the remaining charge originates from the ionized Select alternative (donors, holes, acceptors, electrons). Moving further away from the junction into the neutral p-type material, we find a large number of free Select alternative (holes, electrons, dopants) and Select alternative (neutral, ionized, positive) acceptors. The concentration of charge carriers is Select alternative (larger than, less than, essentially equal to)that of the acceptors.

In the n-type material, there are electrons and Select alternative (ionized, free, neutral, very few) donors. In the neutral region, the Select alternative (entangled, free, immobile) electrons and the Select alternative (positive and free, neutral and free, neutral and immobile, positive and immobile) donors cancel out

with respect to charge. In the
depleted part of the n-type semiconductor, close to the junction, the semiconductor

has a net Select alternative (neutral, positive, negative) charge. Here, the semiconductor is

depleted of Select alternative (carriers, donors, acceptors) and the

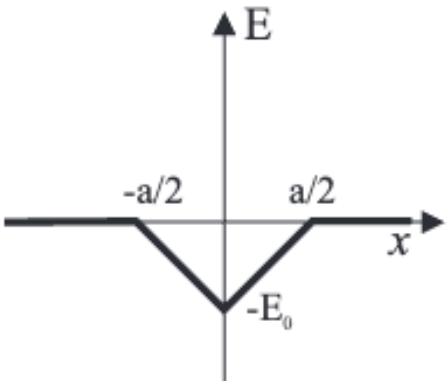
net charge is due to Select alternative (holes, ionized donors, ionized acceptors, unionized

donors). In this particular p-n junction, the doping is highest Select alternative (at the gym, on the p-side, on the n-side, in the middle). As a result, the current at the junction will be dominated by

Select alternative (donors, acceptors, holes, electrons) when the device is under forward bias.

Maximum marks: 14

5 A p-n junction has the E-field profile shown below. Determine the contact potential V_0 and the space charge density $\rho(x)$ as a function of a and E_0.



Space-charge density:
(correct: 4p, wrong: 0p, blank: 0p)
Select one alternative:

- $\rho(x) = \begin{cases} \frac{2\epsilon E_0}{a} \text{ for } x < \frac{-a}{2} \\ 0 \text{ for } \frac{-a}{2} < x < \frac{a}{2} \\ \frac{-2\epsilon E_0}{a} \text{ for } x > \frac{a}{2} \end{cases}$
- $\rho(x) = \begin{cases} 0 \text{ for } x < \frac{-a}{2} \\ \frac{-4\epsilon E_0}{a} \text{ for } \frac{-a}{2} < x < \frac{a}{2} \\ 0 \text{ for } x > \frac{a}{2} \end{cases}$
- $\rho(x) = \begin{cases} 0 \text{ for } x < \frac{-a}{2} \\ \frac{2\epsilon E_0}{a} \text{ for } \frac{-a}{2} < x < 0 \\ \frac{-2\epsilon E_0}{a} \text{ for } 0 < x < \frac{a}{2} \\ 0 \text{ for } x > \frac{a}{2} \end{cases}$
- $\rho(x) = \begin{cases} 0 \text{ for } x < \frac{-a}{2} \\ \frac{-2\epsilon E_0}{a} \text{ for } \frac{-a}{2} < x < 0 \\ \frac{2\epsilon E_0}{a} \text{ for } 0 < x < \frac{a}{2} \\ 0 \text{ for } x > \frac{a}{2} \end{cases}$

Contact potential:
(correct: 4p, wrong: 0p, blank: 0p)

Select one alternative:

- $V_0 = \frac{aE_0}{2}$
- $V_0 = \frac{2E_0}{a}$
- $V_0 = \frac{aE_0}{4}$
- $V_0 = \frac{4E_0}{a}$

Maximum marks: 8

6 *Fill in the blanks in the text below. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)*

A p-n junction is operating under a certain reverse bias. If the reverse bias voltage is increased by a factor 3, what will be the impact on the following parameters?

- The depletion width will Select alternative (Increase, Remain unchanged, Decrease)
- The contact potential will Select alternative (Increase, Decrease, Remain unchanged)
- The junction capacitance will Select alternative (Remain unchanged, Increase, Decrease)
- The probability of breakdown will Select alternative (Decrease, Increase, Remain unchanged)

Maximum marks: 4

7 An abrupt Si junction (area = 0.0001 cm2) has the following parameters:
$N_d = 5 \cdot 10^{17} \mathrm{cm}^{-3}$
$N_a = 1 \cdot 10^{17} \mathrm{cm}^{-3}$

Given: $\epsilon_r = 11.8$

In thermal equilibrium:
What is the built-in potential (contact potential)?
(correct: 4p, wrong: 0p, blank: 0p)

Answer: V

What is the total number of ionized acceptors in the depletion region?
(correct: 4p, wrong: 0p, blank: 0p)

Answer: $\cdot 10^7$

Maximum marks: 8

8 Diode Equation

The current in a p-n junction is described by the diode equation

$$I = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)(e^{qV/kT} - 1) = I_0(e^{qV/kT} - 1)$$

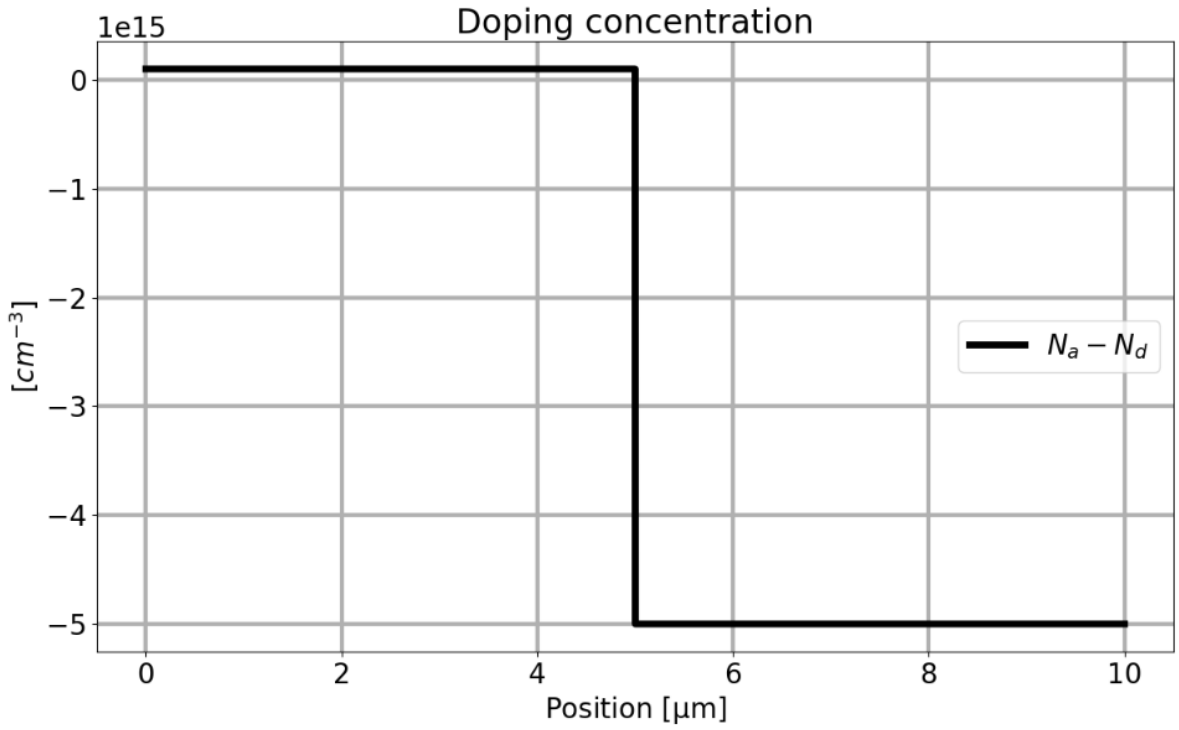
What assumptions have been made in the derivation of this equation?
(correct: 1p, wrong: -1p, blank: 0p, minimum score: 0p)

Select one or more alternatives

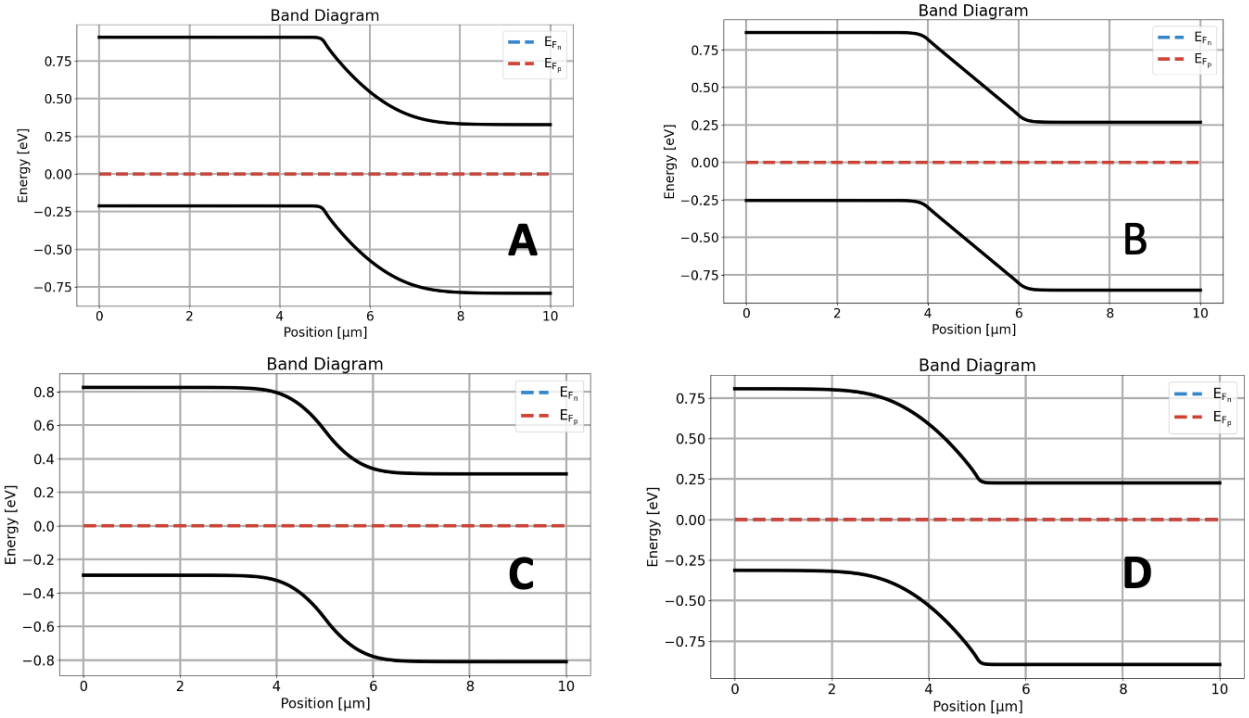
- [ ] Neglect drift of minority carriers
- [ ] Neglect variations in the width of the depletion region with the bias voltage
- [ ] Assume low-level injection
- [ ] Assume abrupt p-n junction
- [ ] Recombination and generation in the transition region (depletion region) is neglected
- [ ] Neglect diffusion of minority carriers
- [ ] Assume high-level injection
- [ ] Neglect voltage drop across the neutral regions

Maximum marks: 4

9 The figure below shows the doping profile $N_a - N_d$ for a p-n junction.



Which of the band diagrams shown below corresponds to a p-n junction with this doping profile?
(correct: 2p, wrong: 0p, blank: 0p)

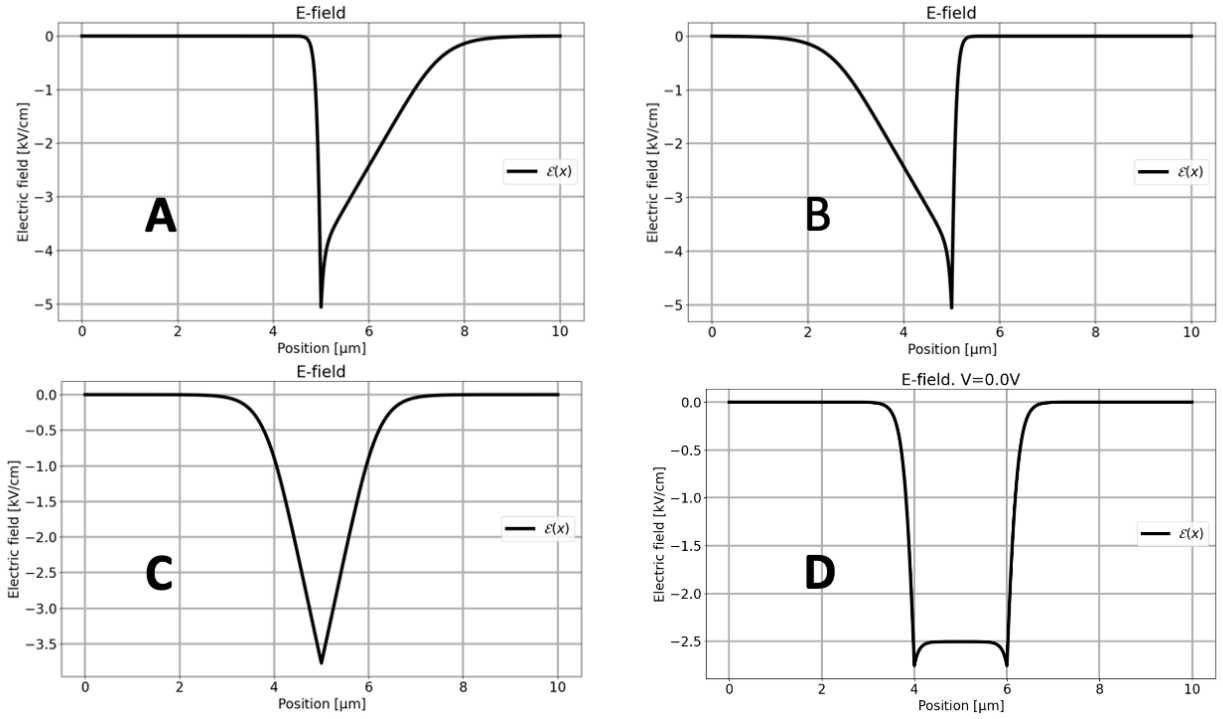


Select one alternative

- C
- B
- A
- D

Which of the electric field profiles shown below corresponds to a p-n junction with this doping profile?

(correct: 2p, wrong: 0p, blank: 0p)

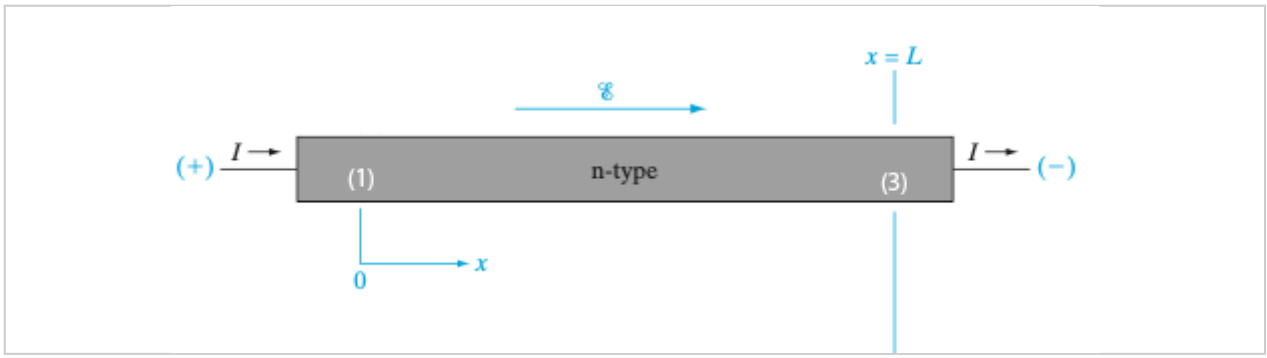


Select one alternative:

- D
- A
- B
- C

Maximum marks: 4

10



Fill in the blanks in the text below. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)

In an n-type bar which contains an electric field, as shown in the figure, the basic principles of the Haynes-Shockley experiment are as follows:

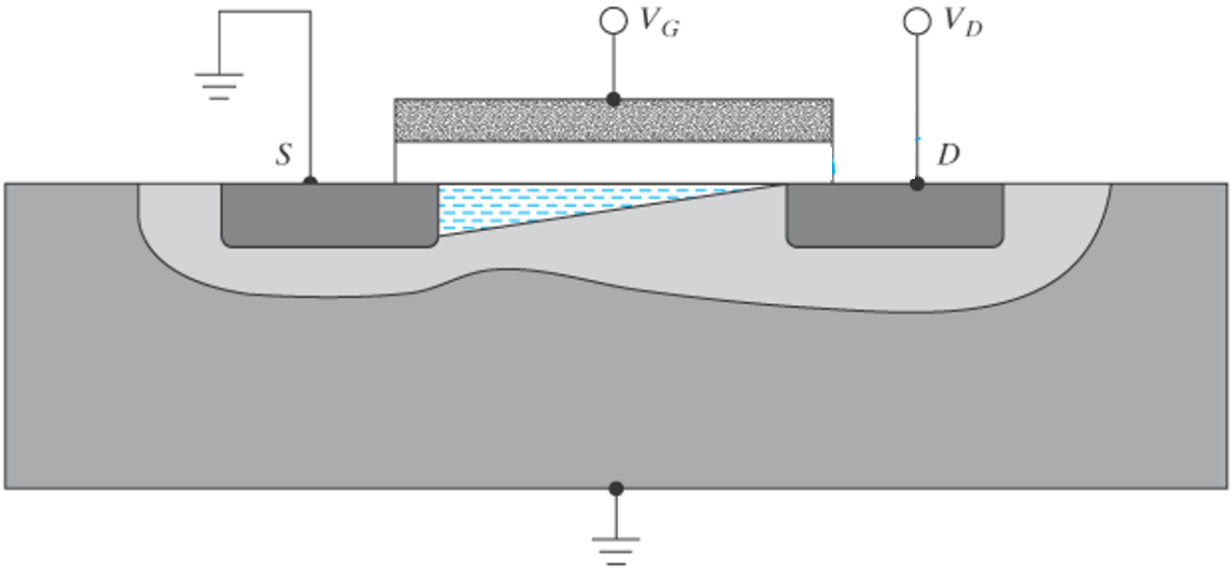
1. A pulse of [Select alternative] (electrons, holes) is created close to one side of the bar (1).

2. As the pulse [Select alternative] (diffuses, drifts) along the bar in the direction of the electric field, it spreads out due to [Select alternative] (diffusion, drift).

3. At the other end of the bar (3), we measure the width of the pulse to calculate the [Select alternative] (carrier type, concentration, diffusion coefficient, mobility)of [Select alternative] (majority carriers, minority carriers). We also measure the drift time of the pulse to calculate the [Select alternative] (mobility, carrier type, diffusion coefficient, concentration) of [Select alternative] (majority carriers, minority carriers).

Maximum marks: 7

11



The figure above shows the cross section of an enhancement mode (normally off) n-channel MOSFET at the onset of saturation.

Based on the indicated depletion region and inversion layer, select the correct voltage relationships.

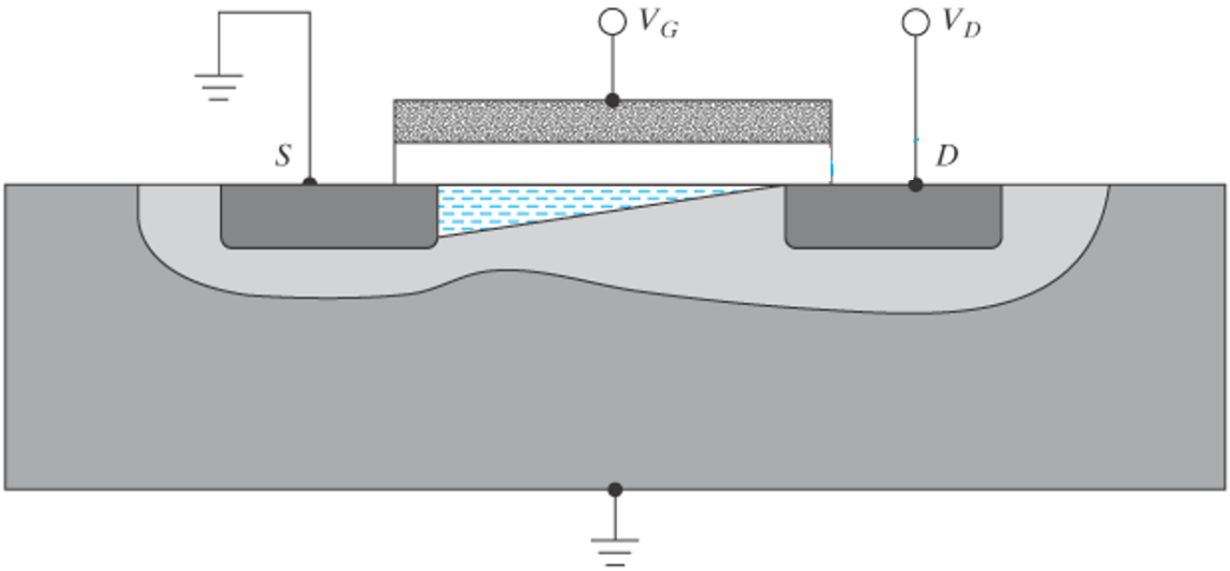
(correct: 2p, wrong: -2p, blank: 0p, minimum score: 0p)

Select one or more alternatives:

- $V_D \approx 0\text{V}$
- $V_G \approx 0\text{V}$
- $V_G > V_T$
- $V_G < V_T$
- $V_D = V_G - V_T$

Maximum marks: 4

12



What will typically be true for a MOSFET with a pinched-off channel?
(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative:

- There is a significant drain current, I_D.
- The device is about to break down.
- The device is in the "OFF" state.
- The drain current is close to zero ($I_D \approx 0$).

Maximum marks: 2

13 The threshold voltage of a MOS transistor is given by $V_T = \Phi_{ms} - \frac{Q_i}{C_i} - \frac{Q_d}{C_i} + 2\phi_F$ **(1)**.

QUESTION 1

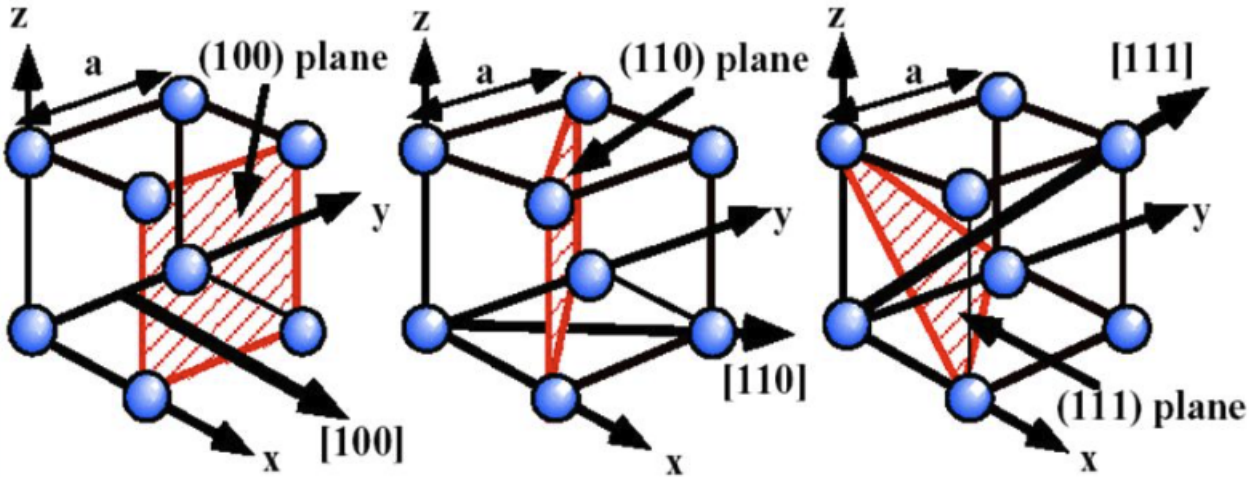
For an NMOS transistor, what will happen to the absolute value of the different terms in **(1)** if the doping concentration in the substrate is increased by a factor 2?

(correct: 2p, wrong: 0p, blank: 0p)

Select one alternative:

- $|\Phi_{ms}|$ increase, $|\phi_F|$ increase, $|Q_d|$ increase
- $|\Phi_{ms}|$ decrease, $|\phi_F|$ increase, $|Q_d|$ increase
- $|\Phi_{ms}|$ increase, $|\phi_F|$ increase, $|Q_d|$ decrease
- $|\Phi_{ms}|$ increase, $|\phi_F|$ decrease, $|Q_d|$ decrease

QUESTION 2



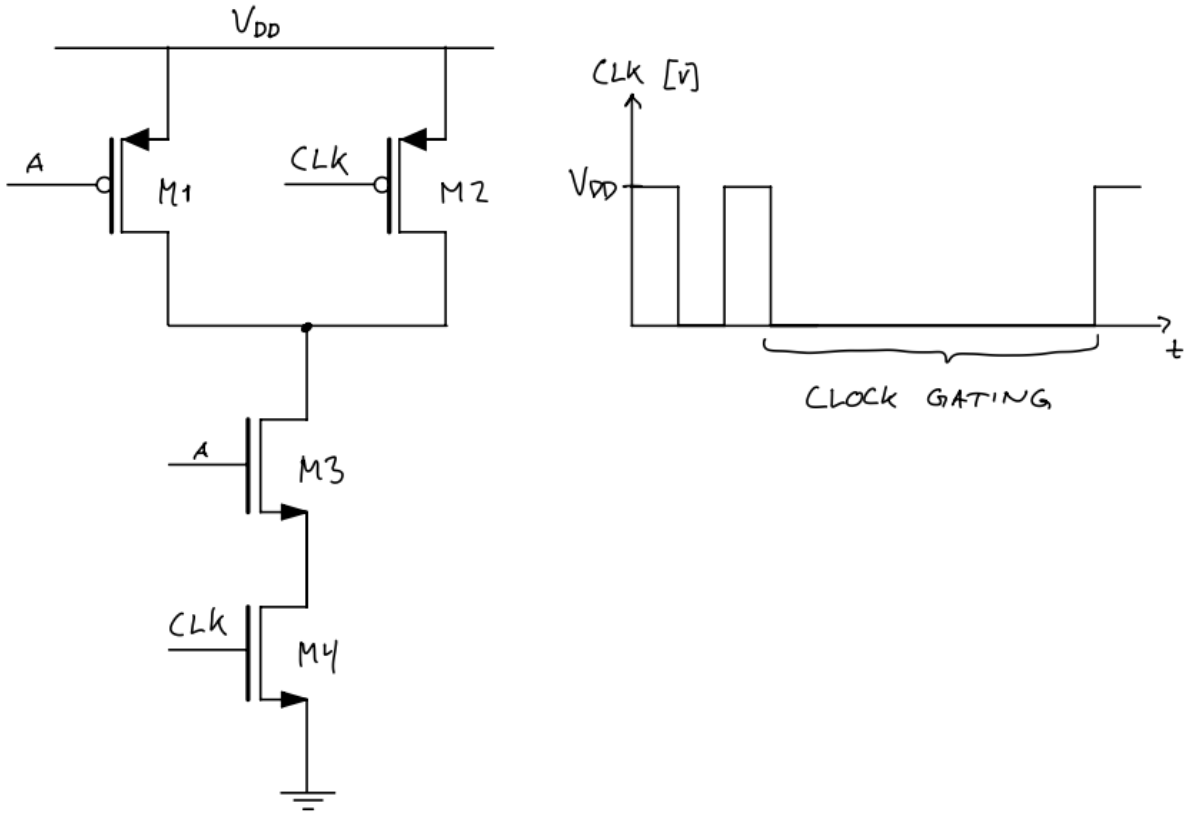
The figure above shows an illustration of a cubic crystal lattice with the {100} plane, {110} plane and the {111} plane indicated. Considering the properties of MOS transistors, what is the main reason why silicon ingots used for fabrication of integrated circuits are grown along the <100> direction?

(correct: 2p, wrong: 0p, blank: 0p)

Select one alternative:

- Large silicon crystals are easiest to grow along the <100> direction, causing a more reliable fabrication process
- With {100} Si, the work function difference is smaller, causing better control of the threshold voltage.
- Using {100} Si results in a lower density of interface charge at the Si-SiO2 interface, which will reduce the threshold voltage of both NMOS and PMOS transistors
- With {100} Si, the direction of the transistor current will be orthogonal to a {110} plane, causing less scattering and higher electron mobility than for {111} Si.

QUESTION 3



Clock gating is a popular technique used to save dynamic power consumption in digital integrated circuits, where the clock signal is disabled for circuits that is not in use. The figure above shows the schematic of a NAND-gate with the two input signals "A" and "CLK". During clock gating, the CLK signal remains at zero volt for a long period of time, and the PMOS transistor M2 will have a strong electric field across the gate oxide. As a consequence, there will be a current of carriers tunneling through the oxide barrier and some of the positive charge carriers will get trapped inside the gate oxide. The concentration of positive trapped charges inside the oxide will increase over time, an effect known as Negative-bias temperature instability.

What could be the consequence of this accumulation of trapped positive charges in the gate oxide?

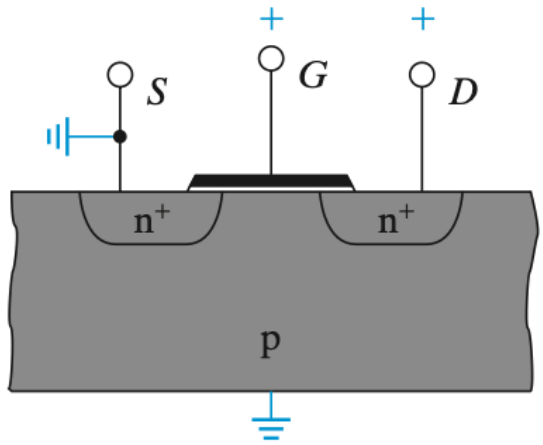
(correct: 2p, wrong: 0p, blank: 0p)

Select one alternative:

- The threshold voltage will increase (decrease in absolute value). This will increase the ON-current of M2, improving the speed of the NAND-gate when the clock is turned back on.
- The electron mobility will increase. This will increase the ON-current of M2, improving the speed of the NAND-gate when the clock is turned back on.
- The hole mobility will decrease. This will reduce the ON-current of M2, such that the NAND gate might no longer meet the timing requirements when the clock is turned back on.
- The threshold voltage will decrease (increase in absolute value). This will reduce the ON-current of M2, such that the NAND gate might no longer meet the timing requirements when the clock is turned back on.

Maximum marks: 6

14



Fill in the blanks in the text below. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)

Consider an NMOS transistor where the source and bulk terminal are connected to grund, as illustrated above. If the gate voltage is gradually increased from zero volt, negative charges will be induced in the underlying Si. For low gate voltages, the induced negative charge is dominated

by Select alternative (mobile electrons, ionized acceptors, ionized donors, electrons in the inversion layer formed under the gate). When the gate voltage is further increased, the

concentration of Select alternative (ionized acceptors, ionized donors, mobile electrons, electron-hole pairs) in a thin surface region under the gate will increase rapidly, forming a conductive channel between the source and drain terminal. When the gate voltage reach

the *threshold voltage* V_T, Select alternative (the current will be essentially unchanged if the gate voltage is increased even further, a significant current will flow through the gate due to quantum tunneling effects., the concentration of mobile electrons in the channel equals the concentration of mobile holes in the substrate, a current can flow between the drain and source

terminal). The transistor is now said to be operating in Select alternative (the linear region, strong inversion, weak inversion, saturation).

If a positive voltage is applied to the drain terminal, a current will flow from drain to source. This

current is dominated by Select alternative (velocity saturation, drift, recombination,

diffusion) of Select alternative (electron-hole pairs, electrons, holes). Because of the positive

drain voltage, the concentration of mobile charge carriers is Select alternative (dominated by

ionized acceptors, highest at the drain-end of the channel, independent of the gate voltage, lowest

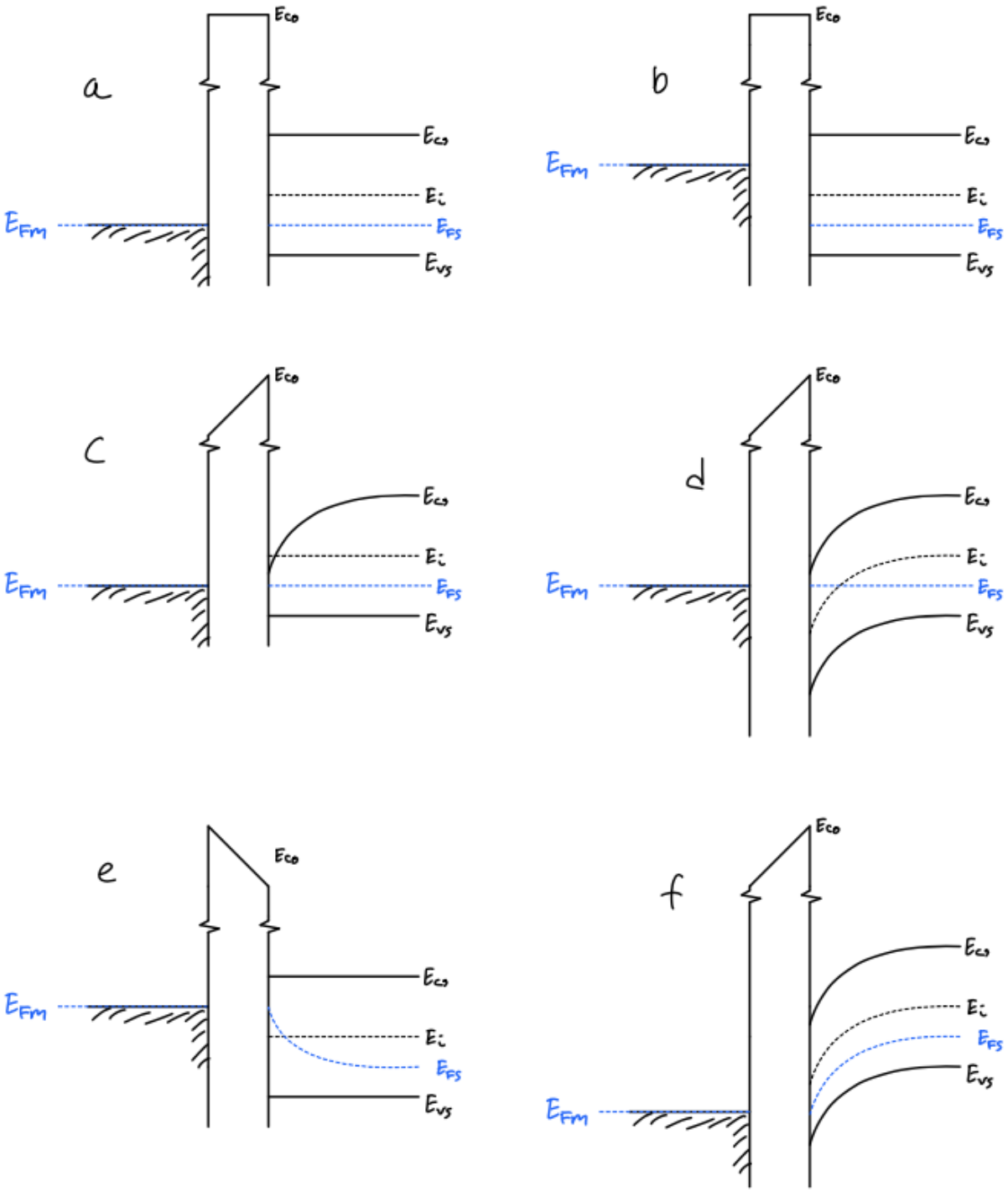
at the drain-end of the channel) and the channel resistance is Select alternative (highest at the source-end of the channel, essentially the same throughout the channel, highest at the drain-end of the channel, zero if the gate voltage is higher than the threshold voltage). The electric field

along the channel is Select alternative (constant throughout the channel, strongest at the source-end of the channel, strongest at the drain-end of the channel, strongest in the middel of

the channel) such that the drain current is Select alternative (zero, constant, highest at the source-end of the channel, highest at the drain-end of the channel).

Maximum marks: 10

15 The figure shows different illustrations of the band diagram of a MOS structure. Which of the illustrations shows a plausible band diagram for a real MOS capacitor?
(correct: 1p, wrong: -1p, blank: 0p, minimum score: 0p)

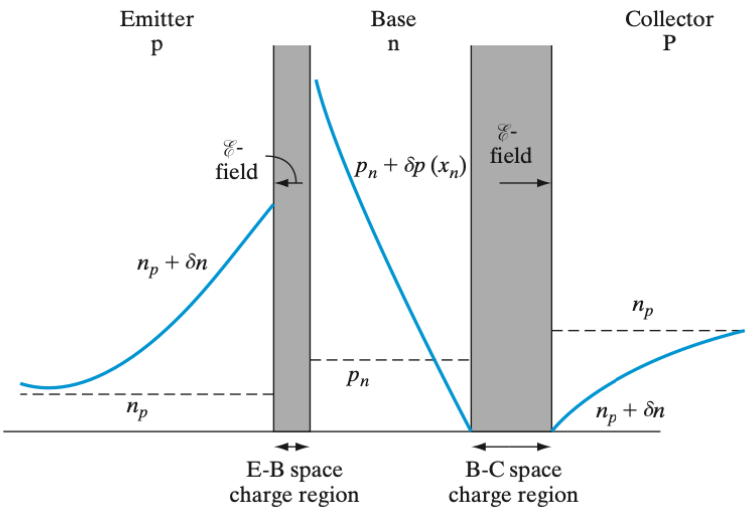


Select one or more alternatives:

- [ ] d
- [ ] f
- [ ] e
- [ ] a
- [ ] c
- [ ] b

Maximum marks: 3

16



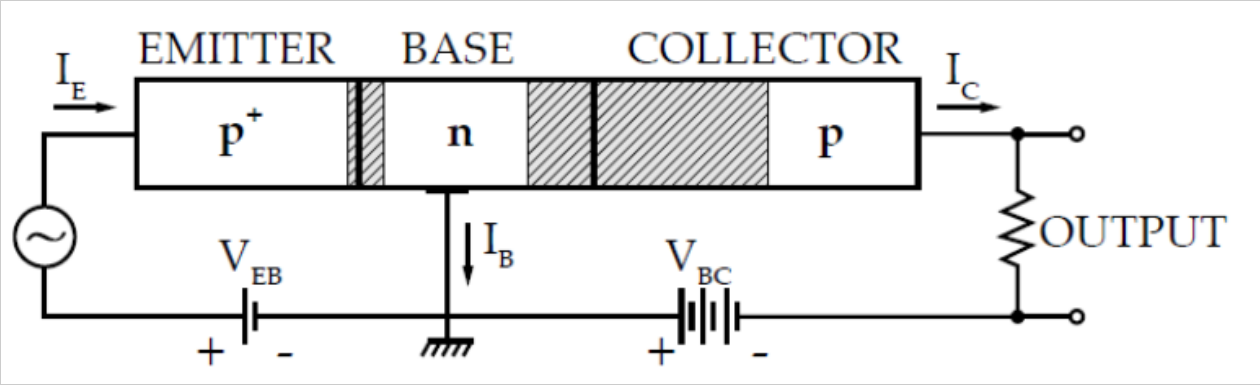
The figure above shows the minority carrier distributions of a PNP bipolar junction transistor (BJT) in the normal active mode. The hole concentration at the emitter- and collector-end of the base region is Δp_E and Δp_C, respectively, the width of the neutral base region is W_b, the cross-section area of the BJT is A and the minority carrier lifetime in the base is τ_p.

Assuming $\delta p_n(x_n)$ is linear in the base region, what is the expression for the base current?
(correct: 4p, wrong: 0p, blank: 0p)
Select one alternative:

- $I_B = qA\frac{D_p}{L_p}\left[(\Delta p_E + \Delta p_C)\tanh\frac{W_b}{2L_p}\right]$
- $I_B = \frac{qA\Delta p_E W_b}{2\tau_p}$
- $I_B = \frac{qAL_p\Delta p_n}{\tau_p}$
- $I_B = \frac{qA\Delta p_E W_b}{\tau_p}$

Maximum marks: 4

17



For the BJT shown above, which relationship between the doping concentrations agrees with the sketch:
(correct: 2p, wrong: 0p, blank: 0p)

- $N_E < N_B < N_C$
- $N_E > N_B > N_C$
- $N_E < N_B > N_C$
- $N_E > N_B < N_C$

What is the operation mode of the BJT shown above:
(correct: 2p, wrong: 0p, blank: 0p)

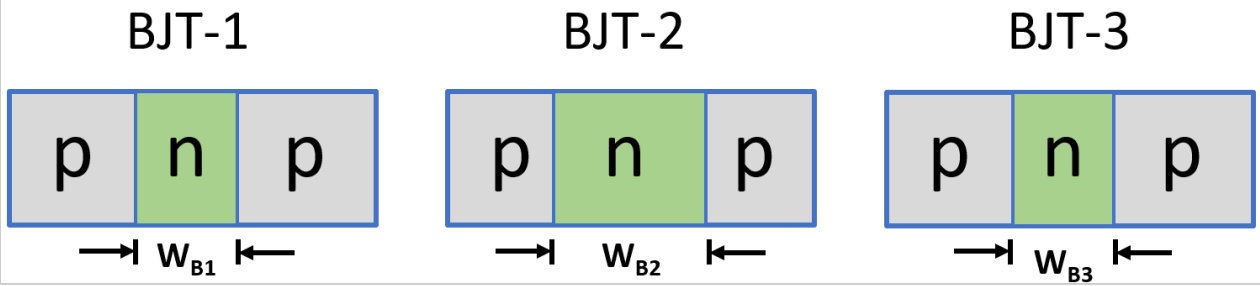
- Cutoff
- Normal active
- Inverted
- Saturation

What is the the biasing arrangement shown above called:
(correct: 2p, wrong: 0p, blank: 0p)

- Gummel-Poon
- Common emitter
- Coupled-diode
- Common base

Maximum marks: 6

18



The three p-n-p bipolar junction transistors schematically illustrated above, are identical except that BJT-2 has a base region that is twice as wide as that of BJT-1 and BJT-3. We also have that BJT-3 and BJT-2 have a doping in the base region that is twice as large as that of BJT-1. That is:

$W_{B2} = 2W_{B1} = 2W_{B3}$ and $N_{B3} = N_{B2} = 2N_{B1}$

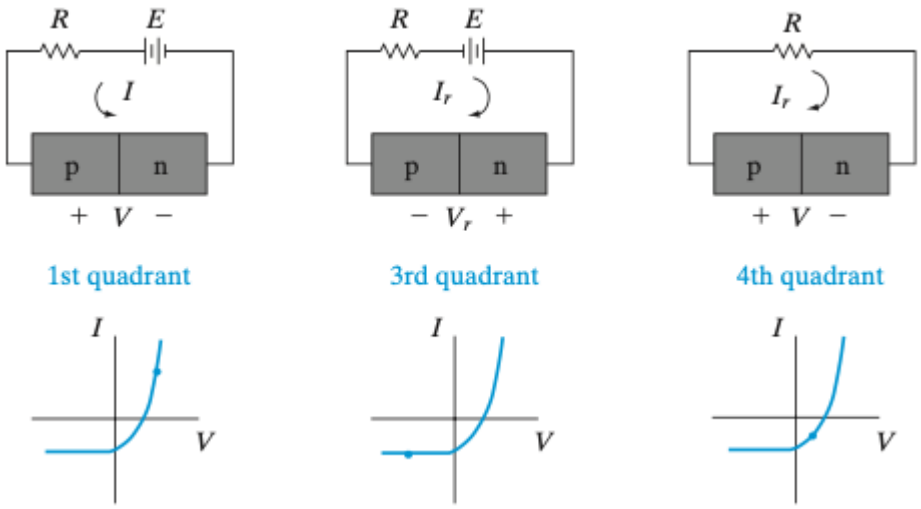
We assume that $N_E > N_B > N_C$ for all three transistors.
(correct: 1p, wrong: -1p, blank: -1p, minimum score: 0p)

Which of the three transistors will have the highest value for:

	BJT-3	BJT-1	BJT-2
Common emitter gain	○	○	○
"Punch-through" voltage	○	○	○
Base transport factor	○	○	○
Emitter injection efficiency	○	○	○

Maximum marks: 4

19 The figure below shows a p-n junction operated in the various quadrants of its I-V characteristic.



(correct: 1p, wrong: -1p, blank: -1p, minimum score 0p)

Select which quadrant the junction should be operated in, when it is to be used as a solar cell, photodetector and a LASER.

	4th quadrant	1st quadrant	3rd quadrant
Photodetector	○	○	○
Solar cell	○	○	○
LASER	○	○	○

Maximum marks: 3

20

InsperaScan (30 p.)

Problem a-e gives 6p each

Maximum marks: 30

ℹ Formulas

ℹ Constants and conversions

Electrostatic discharge

Electrostatic discharge (ESD) is the phenomena where an electrostatic voltage spontaneously discharges as an electrical current impulse. In a typical work environment, a charge of about 0.6µC can be induced on a body capacitance of 150µF, leading to electrostatic potentials of 4000 V or greater. Any contact by the charged human body with a grounded object, such as an integrated circuit (IC) pin, can result in discharge currents of several amperes, potentially causing significant damage to the IC. In the design of ICs, it is therefore common practice to include an ESD protection circuit, protecting the internal circuits from ESD currents.

A typical context of an ESD protection circuit is shown if figure 1. A pin is used to interface with an integrated circuit. When a normal signal, e.g. a logical high voltage, is applied to the pin, we want the current to flow into the circuit. However, in the case of an ESD event, we want the high ESD current to flow directly to ground through the ESD protection circuit, thereby leaving the circuit undamaged.

A standard solution to this problem, is the so-called grounded-gate NMOS (GGNMOS), shown in figure 2. A sketch of the cross-section is shown in figure 3.

a) Draw a similar cross-section and indicate the border of the depletion regions when the drain voltage (i.e. the pin voltage), V_D, is

   i) 0V

   ii) 1.5V

b) When the PIN voltage is 1.5V, only a small leakage current will flow to ground through the drain-bulk junction of the GGNMOS.

   i) Give an expression for the leakage current density (current per area).

   ii) Will the current through the drain-bulk junction be dominated by holes or electrons?

   iii) Show that the leakage current will increase with increasing temperature.

c) When an ESD event occurs, the PIN voltage of 1000V will cause a reverse brakedown in the drain-bulk junction. Will this be a Zener brakedown or an avalanche breakdown? Justify your answer.

d) Calculate the conductivity in the substrate when the doping concentration is $N_A = 10^{14}\text{cm}^{-3}$. Assume $\mu_n = 1350\text{cm}^2/\text{Vs}$ and $\mu_p = 480\text{cm}^2/\text{Vs}$.

e) The breakdown current will flow through the substrate to the p^+ region, which forms an ohmic contact to a metal wire connected to ground. Due to the finite conductivity in the lightly p-doped substrate, there will be an increased potential in the substrate, which will cause the source-bulk junction to become forward biased. This will "turn on" the parasitic NPN bipolar transistor consisting of the source (emitter), substrate (base) and drain (collector) regions.

   Sketch and explain the flow of electrons and holes in this NPN transistor when it operates in the normal active mode. Let the thickness of the arrows indicate the relative size of the current components.

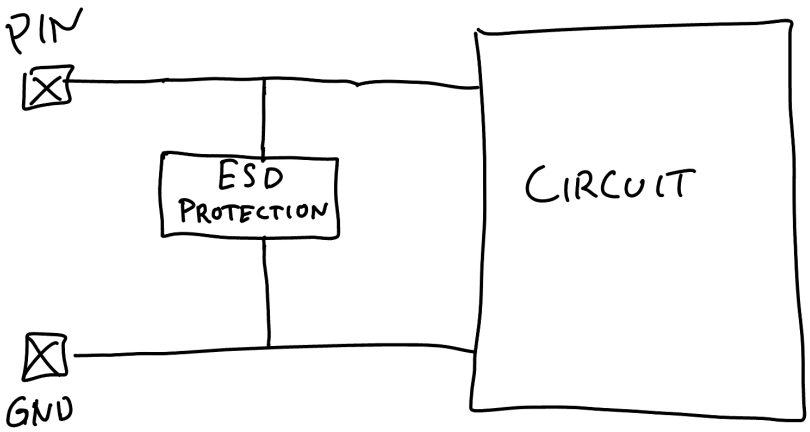


Figure 1: The context of an ESD protection circuit

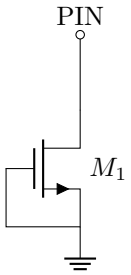


Figure 2: Circuit schematic of the grounded gate NMOS

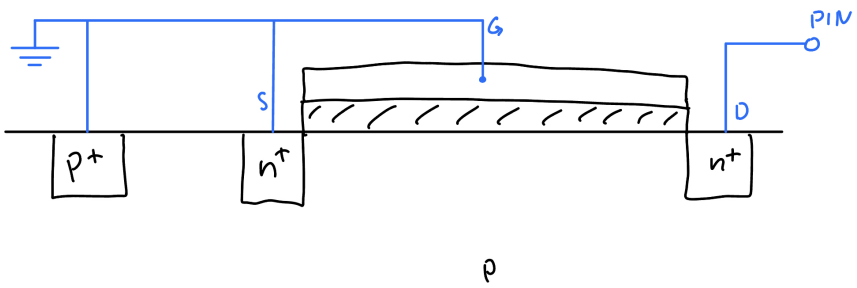


Figure 3: Sketch of the cross-section of a GGNMOS.

Formulas TFE-4146

SEMICONDUCTOR PHYSICS

Electron Momentum: $\mathrm{p} = m^*\mathrm{v} = \hbar\mathbf{k} = \dfrac{h}{\lambda}$ Planck: $E = h\nu = \hbar\omega$

Kinetic: $E = \dfrac{1}{2}m^*\mathrm{v}^2 = \dfrac{1}{2}\dfrac{\mathrm{p}^2}{m^*} = \dfrac{\hbar^2}{2m^*}\mathbf{k}^2$ (3-4) Effective mass: $m^* = \dfrac{\hbar^2}{d^2E/d\mathbf{k}^2}$ (3-3)

Total electron energy = $P.E. + K.E. = E_c + E(\mathbf{k})$

Fermi-Dirac e^- distribution: $f(E) = \dfrac{1}{e^{(E-E_F)/kT}+1} \cong e^{(E_F-E)/kT}$ for $E \gg E_F$ (3-10)

Equilibrium: $n_0 = \displaystyle\int_{E_c}^{\infty} f(E)N(E)dE = N_c f(E_c) = N_c e^{-(E_c-E_F)/kT}$ (3-15)

$N_c = 2\left(\dfrac{2\pi m_n^* kT}{h^2}\right)^{3/2}$ $N_v = 2\left(\dfrac{2\pi m_p^* kT}{h^2}\right)^{3/2}$ (3-16), (3-20)

$p_0 = N_v[1 - f(E_v)] = N_v e^{-(E_F-E_v)/kT}$ (3-19)

$n_i = N_c e^{-(E_c-E_i)/kT}$, $p_i = N_v e^{-(E_i-E_v)/kT}$ (3-21)

$n_i = \sqrt{N_c N_v}\, e^{-E_g/2kT} = 2\left(\dfrac{2\pi kT}{h^2}\right)^{3/2}(m_n^* m_p^*)^{3/4}e^{-E_g/2kT}$ (3-23), (3-26)

Equilibrium: $n_0 = n_i e^{(E_F-E_i)/kT}$, $p_0 = n_i e^{(E_i-E_F)/kT}$ (3-25) $n_0 p_0 = n_i^2$ (3-24)

Steady state: $n = N_c e^{-(E_c-F_n)/kT} = n_i e^{(F_n-E_i)/kT}$, $p = N_v e^{-(F_p-E_v)/kT} = n_i e^{(E_i-F_p)/kT}$ (4-15) $np = n_i^2 e^{(F_n-F_p)/kT}$ (5-38)

$\mathscr{E}(x) = -\dfrac{d\mathscr{V}(x)}{dx} = \dfrac{1}{q}\dfrac{dE_i}{dx}$ (4-26)

Poisson: $\dfrac{d\mathscr{E}(x)}{dx} = -\dfrac{d^2\mathscr{V}(x)}{dx^2} = \dfrac{\rho(x)}{\epsilon} = \dfrac{q}{\epsilon}(p - n + N_d^+ - N_a^-)$ (5-14)

$\mu \equiv \dfrac{q\bar{t}}{m^*}$ (3-40a) Drift: $\mathrm{v}_d \cong \dfrac{\mu\mathscr{E}}{1 + \mu\mathscr{E}/\mathrm{v}_s}\begin{cases} = \mu\mathscr{E} \text{ (low fields, ohmic)} \\ = \mathrm{v}_s \text{ (high fields, saturated vel.)} \end{cases}$ (Fig. 6-9)

Drift current density: $\dfrac{I_x}{A} = J_x = q(n\mu_n + p\mu_p)\mathscr{E}_x = \sigma\mathscr{E}_x$ (3-43)

$$J_n(x) = q\mu_n n(x)\mathscr{E}(x) + qD_n \frac{dn(x)}{dx}$$

Conduction Current: drift diffusion (4–23)

$$J_p(x) = q\mu_p p(x)\mathscr{E}(x) - qD_p \frac{dp(x)}{dx}$$

$$J_{\text{total}} = J_{\text{conduction}} + J_{\text{displacement}} = J_n + J_p + C\frac{dV}{dt}$$

Continuity: $\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$ $\frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}$ (4–31)

For steady state diffusion: $\frac{d^2\delta n}{dx^2} = \frac{\delta n}{D_n\tau_n} \equiv \frac{\delta n}{L_n^2}$ $\frac{d^2\delta p}{dx^2} = \frac{\delta p}{L_p^2}$ (4–34)

Diffusion length: $L \equiv \sqrt{D\tau}$ Einstein relation: $\frac{D}{\mu} = \frac{kT}{q}$ (4–29)

p-n JUNCTIONS

Equilibrium: $V_0 = \frac{kT}{q}\ln\frac{p_p}{p_n} = \frac{\text{kT}}{\text{q}}\ln\frac{N_a}{n_i^2/N_d} = \frac{\text{kT}}{\text{q}}\ln\frac{N_aN_d}{n_i^2}$ (5–8)

$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_0/kT}$ (5–10) $W = \left[\frac{2\epsilon(V_0 - V)}{q}\left(\frac{N_a + N_d}{N_aN_d}\right)\right]^{1/2}$ (5–57)

One-sided abrupt p^+-n: $x_{n0} = \frac{WN_a}{N_a + N_d} \simeq W$ (5–23) $V_0 = \frac{qN_dW^2}{2\epsilon}$

$\Delta p_n = p(x_{n0}) - p_n = p_n(e^{qV/kT} - 1)$ (5–29)

$\delta p(x_n) = \Delta p_n e^{-x_n/L_p} = p_n(e^{qV/kT} - 1)e^{-x_n/L_p}$ (5–31b)

Ideal diode: $I = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)(e^{qV/kT} - 1) = I_0(e^{qV/kT} - 1)$ (5–36)

Non-ideal: $I = I_0'(e^{qV/\mathfrak{n}kT} - 1)$ ($\mathfrak{n}$ = 1 to 2) (5–74)

With light: $I_{\text{op}} = qAg_{\text{op}}(L_p + L_n + W)$ (8–1)

$$I_D = \frac{\bar{\mu}_n Z C_i}{L}\left\{(V_G - V_{FB} - 2\phi_F - \tfrac{1}{2}V_D)V_D - \frac{2}{3}\frac{\sqrt{2\epsilon_s q N_a}}{C_i}[(V_D + 2\phi_F)^{3/2} - (2\phi_F)^{3/2}]\right\} \quad (6\text{-}50)$$

$$I_D \simeq \frac{\bar{\mu}_n Z C_i}{L}[(V_G - V_T)V_D - \tfrac{1}{2}V_D^2] \quad (6\text{-}49)$$

Saturation: $I_D(\text{sat.}) \simeq \frac{1}{2}\bar{\mu}_n C_i \frac{Z}{L}(V_G - V_T)^2 = \frac{Z}{2L}\bar{\mu}_n C_i V_D^2(\text{sat.}) \quad (6\text{-}53)$

$$g_m = \frac{\partial I_D}{\partial V_G} \quad ; \quad g_m(\text{sat.}) = \frac{\partial I_D(\text{sat.})}{\partial V_G} \simeq \frac{Z}{L}\bar{\mu}_n C_i(V_G - V_T) \quad (6\text{-}54)$$

For short L: $I_D \simeq ZC_i(V_G - V_T)v_s \quad (6\text{-}60)$

Subthreshold slope: $S = \frac{dV_G}{d(\log I_D)} = \frac{kT}{q}\ln 10\left[1 + \frac{C_d + C_{it}}{C_i}\right] \quad (6\text{-}66)$

BJT-p-n-p

$$I_{Ep} = qA\frac{D_p}{L_p}\left(\Delta p_E \operatorname{ctnh}\frac{W_b}{L_p} - \Delta p_C \operatorname{csch}\frac{W_b}{L_p}\right) \quad (7\text{-}18) \qquad \begin{aligned}\Delta p_E &= p_n(e^{qV_{EB}/kT} - 1)\\ \Delta p_C &= p_n(e^{qV_{CB}/kT} - 1)\end{aligned} \quad (7\text{-}8)$$

$$I_C = qA\frac{D_p}{L_p}\left(\Delta p_E \operatorname{csch}\frac{W_b}{L_p} - \Delta p_C \operatorname{ctnh}\frac{W_b}{L_p}\right)$$

$$I_B = qA\frac{D_p}{L_p}\left[(\Delta p_E + \Delta p_C)\tanh\frac{W_b}{2L_p}\right] \quad (7\text{-}19)$$

$$B = \frac{I_C}{I_{Ep}} = \frac{\operatorname{csch} W_b/L_p}{\operatorname{ctnh} W_b/L_p} = \operatorname{sech}\frac{W_b}{L_p} \simeq 1 - \left(\frac{W_b^2}{2L_p^2}\right) \quad (7\text{-}26)$$

(Base transport factor)

$$\gamma = \frac{I_{Ep}}{I_{En} + I_{Ep}} = \left[1 + \frac{L_p^n n_n \mu_n^p}{L_n^p p_p \mu_p^n}\tanh\frac{W_b}{L_p^n}\right]^{-1} \simeq \left[1 + \frac{W_b n_n \mu_n^p}{L_n^p p_p \mu_p^n}\right]^{-1} \quad (7\text{-}25)$$

(Emitter injection efficiency)

$$\frac{i_C}{i_E} = B\gamma \equiv \alpha \quad (7\text{-}3) \qquad \frac{i_C}{i_B} = \frac{B\gamma}{1 - B\gamma} = \frac{\alpha}{1 - \alpha} \equiv \beta \quad (7\text{-}6) \qquad \frac{i_C}{i_B} = \beta = \frac{\tau_p}{\tau_t} \quad (7\text{-}7)$$

(Common base gain) (Common emitter gain) (For $\gamma = 1$)

Capacitance: $C = \left|\frac{dQ}{dV}\right|$ (5-55)

Junction Depletion: $C_j = \epsilon A\left[\frac{q}{2\epsilon(V_0 - V)}\frac{N_d N_a}{N_d + N_a}\right]^{1/2} = \frac{\epsilon A}{W}$ (5-62)

Stored charge
exp. hole dist.: $Q_p = qA\int_0^\infty \delta p(x_n)dx_n = qA\Delta p_n\int_0^\infty e^{-x_n/L_p}dx_n = qAL_p\Delta p_n$ (5-39)

$I_p(x_n = 0) = \frac{Q_p}{\tau_p} = qA\frac{L_p}{\tau_p}\Delta p_n = qA\frac{D_p}{L_p}p_n(e^{qV/kT} - 1)$ (5-40)

$G_s = \frac{dI}{dV} = \frac{qAL_p p_n}{\tau_p}\frac{d}{dV}(e^{qV/kT}) = \frac{q}{kT}I$ (5-67c)

Long p^+-n: $i(t) = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt}$ (5-47)

MOS-n CHANNEL

Oxide: $C_i = \frac{\epsilon_i}{d}$ Depletion: $C_d = \frac{\epsilon_s}{W}$ MOS: $C = \frac{C_i C_d}{C_i + C_d}$ (6-36)

Threshold: $V_T = \underbrace{\Phi_{ms} - \frac{Q_i}{C_i}}_{\text{Flat band}} - \frac{Q_d}{C_i} + 2\phi_F$ (6-38)

Inversion: $\phi_s\,(\text{inv.}) = 2\phi_F = 2\frac{kT}{q}\ln\frac{N_a}{n_i}$ (6-15) $W = \left[\frac{2\epsilon_s\phi_s}{qN_a}\right]^{1/2}$ (6-30)

$Q_d = -qN_aW_m = -2(\epsilon_s qN_a\phi_F)^{1/2}$ (6-32) At V_{FB}: $C_{FB} = \frac{C_i C_{\text{debye}}}{C_i + C_{\text{debye}}}$

Debye screening length: $L_D = \sqrt{\frac{\epsilon_s kT}{q^2 p_0}}$ (6-25) $C_{\text{debye}} = \frac{\epsilon_s}{L_D}$ (6-40)

Substrate bias: $\Delta V_T \simeq \frac{\sqrt{2\epsilon_s qN_a}}{C_i}(-V_B)^{1/2}$ (n channel) (6-63)

Physical Constants and Conversion Factors for the Exam in TFE-4146

Avogadro's number	$N_A = 6.02 \times 10^{23}$ molecules/mole
Boltzmann's constant	$k = 1.38 \times 10^{-23}$ J/K
	$= 8.62 \times 10^{-5}$ eV/K
Electronic charge (magnitude)	$q = 1.60 \times 10^{-19}$ C
Electronic rest mass	$m_0 = 9.11 \times 10^{-31}$ kg
Permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-14}$ F/cm
	$= 8.85 \times 10^{-12}$ F/m
Planck's constant	$h = 6.63 \times 10^{-34}$ J-s
	$= 4.14 \times 10^{-15}$ eV-s
Room temperature value of kT	$kT = 0.0259$ eV
Speed of light	$c = 2.998 \times 10^{10}$ cm/s

1 Å (angstrom) $= 10^{-8}$ cm	
1 μm (micron) $= 10^{-4}$ cm	
1 nm = 10 Å $= 10^{-7}$ cm	
2.54 cm = 1 in.	
1 eV $= 1.6 \times 10^{-19}$ J	

Prefixes:

milli-,	m-	$= 10^{-3}$
micro-,	μ-	$= 10^{-6}$
nano-,	n-	$= 10^{-9}$
pico-,	p-	$= 10^{-12}$
kilo-,	k-	$= 10^{3}$
mega-,	M-	$= 10^{6}$
giga-,	G-	$= 10^{9}$

A wavelength λ of 1 μm corresponds to a photon energy of 1.24 eV.